

# AMD Ryzen 7 9850X3D

## Polishing the X3D Formula to a Shine

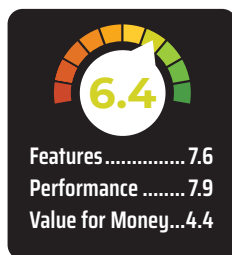
**W**ith the AMD Ryzen 7 9850X3D, the company is essentially doing something it has been doing for a while now, iterate from a position of comfort. The Ryzen 7 9800X3D was already a bit of a cheat code for high-refresh gaming, so the 9850X3D's job is not to reinvent the formula, it's to sand down the rough edges and squeeze a little more out of the same basic idea. And on paper, that's exactly what AMD has done: the same 8-core, 16-thread Zen 5 setup, the same X3D cache configuration, the same 120W class power envelope, but with higher boost clocks and what AMD claims are small, repeatable gains in the kinds of games that are sensitive to frequency and frame-time consistency.

If that sounds familiar, it should. This while packaged as a new tier of performance that forces everyone else to respond overnight, is actually a refinement aimed at enthusiasts who already know why X3D chips matter. The bigger question is whether the uplift is meaningful enough to justify its place next to the 9800X3D, and whether its pricing and availability in India land in the right spot.

### PERFORMANCE

The Ryzen 7 9850X3D is built for one job first: gaming. The higher boost clock is the main lever, so most of the uplift shows up where games are CPU-limited, frame rates are high, and small frequency shifts can translate into better 1% lows and slightly higher averages.

Outside gaming, the performance is subtle. With the same core count as the 9800X3D, multi-threaded heavy lifting tends to scale similarly, with modest improvements in select workloads thanks to the higher clocks and tuning



refinements rather than any big architectural jump.

### Cinebench 2024

Cinebench 2024 is based on Maxon's Cinema 4D software, designed to evaluate a processor's performance in rendering complex 3D

scenes. It tests both single-core and multi-core capabilities, highlighting how efficiently a CPU handles multi-threaded tasks common in professional rendering workflows. The benchmark utilises modern instruction sets and large datasets, providing an up-to-date assessment of CPU performance in contemporary applications.

### Cinebench R23

Cinebench R23 is the older version based on the same Cinema 4D engine and measures a processor's ability to render photorealistic 3D scenes. Replaced by Cinebench 2024, it's useful for comparing against some older processors. It focuses on both single and multi-threaded performance, simulating real-world tasks relevant to content creators and professionals.

### Blender

The Blender Benchmark assesses a processor's performance by rendering scenes using Blender, a popular open-source 3D creation suite. It evaluates how well the CPU handles complex rendering tasks and multi-threaded workloads. For an 8-core X3D chip like the 9850X3D, results tend to be strong for its class, but higher-core-count CPUs will still pull ahead when raw thread count is the deciding factor.

### V-Ray

V-Ray is another benchmark that also measures a processor's performance in rendering but it uses the V-Ray engine, widely used in visual effects, architecture, and the design industries. It tests multi-threaded performance and handling complex calculations that are involved in ray tracing. CPUs with higher core counts tend to shine, but the 9850X3D's higher clocks help keep it competitive within the 8-core bracket.

### WinRAR

WinRAR Benchmark evaluates a processor's performance in data compression and decompression tasks, common in file archiving and management. It leans on single-threaded performance, memory bandwidth, and latency. With the 9850X3D, expectations should be modest uplift versus the 9800X3D rather than a new class of result, because core count and cache layout remain unchanged.

### AIDA64

AIDA64 Memory Benchmark measures the memory bandwidth and latency of a system, highlighting how quickly data transfers between the CPU and RAM. It tests read, write, and copy speeds, providing insights into the efficiency of the memory subsystem, including RAM speed, timings, and memory controller performance. AIDA64 also has benchmarks to evaluate how well a processor can handle AES, ZLib and SHA3 encryption and decryption tasks.